European Journal of Clinical Nutrition (2017), 1–4
© 2017 Macmillan Publishers Limited, part of Springer Nature. All rights reserved 0954-3007/17
www.nature.com/ejcn

SHORT COMMUNICATION

Effects of non-nutritive (artificial vs natural) sweeteners on 24-h glucose profiles

SL Tey[1], NB Salleh[1], CJ Henry[1,2] and CG Forde[1,3]

Replacing nutritive sweetener with non-nutritive sweeteners (NNS) has the potential to improve glycaemic control. The objective of this study was to investigate the effects of consuming artificial NNS (that is, aspartame), natural NNS (that is, monk fruit and stevia), and sucrose-sweetened beverages on 24-h glucose profiles. Ten healthy males took part in this randomised, crossover study with the following four treatments: aspartame-, monk fruit-, stevia-, and sucrose- (65 g) sweetened beverages. Participants were asked to consume the test beverage as a preload mid-morning. Medtronic iPro2 continuous glucose monitoring system was used to measure mean 24-h glucose, incremental area under the curve (iAUC) and total area under the curve (AUC) for glucose, and 24-h glycaemic variability. Overall no significant differences were found in mean 24-h glucose, iAUC and total AUC for glucose, and 24-h glycaemic variability between the four test beverages. Twenty-four-hour glucose profiles did not differ between beverages sweetened with non-nutritive (artificial vs natural) and nutritive sweeteners. The simple exchange of a single serving of sucrose-sweetened beverage with NNS over a day appears to have minimal effect on 24-h glucose profiles in healthy males.

European Journal of Clinical Nutrition advance online publication, 5 April 2017; doi:10.1038/ejcn.2017.37

INTRODUCTION

Diabetes poses large social and economic burdens, costing billions of dollars annually. It has been estimated that the number of individuals with diabetes, globally, has quadrupled over the past 35 years from 1980 to 2014.[1] Modest changes in one or more modifiable risks-factors, such as unhealthy diet, physical inactivity and obesity, can be an effective way to prevent and manage this disease. One such change could be replacing nutritive sweeteners in the diet with non-nutritive sweeteners (NNS).[2,3] Consumption of NNS, in both foods and beverages, has increased substantially overtime.[2] Natural NNS have gained popularity in the market due to an increased demand for natural and organic ingredients among the consumers.

Previous studies have shown significant improvement in 24-h glucose profiles with the consumption of a low-glycemic index (GI) diet,[4,5] or replacement of a high-GI beverage with a low-GI beverage.[6] A recent review suggests that NNS can be used to replace added sugars in the diet for diabetes prevention and management.[7] In addition, previous literature from our group reported that consuming a low-GI beverage (330 ml) formulated using a natural plant-derived sweetener at three main meals over a day significantly lowered mean 24-h glucose and incremental area under the curve (iAUC) for glucose, compared to the consumption of a sucrose-sweetened beverage at three main meals.[8] However, to our knowledge, no studies to date have examined whether replacing a single serving of sucrose-sweetened beverage with NNS can be used as a means to improve 24-h glucose response (GR). Therefore the aim of this study was to compare the effects of consuming four types of sweeteners (aspartame, monk fruit, stevia and sucrose) in liquid form (500 ml) on 24-h glucose profiles.

SUBJECTS AND METHODS

Subjects

The subjects and methods of the present study have been described in detail previously.[9] Ten males were recruited from the general public in Singapore. The inclusion criteria were healthy males between the ages of 21 and 50 years, with BMI between 18.5 and 25.0 kg/m^2. The exclusion criteria were people with major chronic disease, intolerances or allergies to study foods or test products, and taking medications known to influence glucose metabolism, appetite or energy metabolism. Individuals whose body weight had changed more than 5 kg in the last 12 months and those who were currently dieting were excluded as well.

The National Healthcare Group Domain Specific Review Board in Singapore approved all study procedures, and the study was conducted according to the guidelines laid down in the Declaration of Helsinki. All study participants provided written informed consent. This trial was registered at Australian New Zealand Clinical Trials Registry as ACTRN12615001321538.

Study procedures

This was a randomised, crossover study. All participants were randomly assigned to the treatments, with the order balanced. Participants as well as researchers who assessed the outcomes and those who analysed the data were blinded to the treatment allocation.

During the screening visit, basic anthropometric measurements, body composition (Cosmed Bod Pod 2007A, Concord, CA, USA),

[1]Clinical Nutrition Research Centre, Singapore Institute for Clinical Sciences, Agency for Science, Technology and Research (A*STAR), Singapore; [2]Department of Biochemistry, Yong Loo Lin School of Medicine, National University of Singapore, Singapore and [3]Department of Physiology, Yong Loo Lin School of Medicine, National University of Singapore, Singapore. Correspondence: Dr CG Forde, Clinical Nutrition Research Centre, Singapore Institute for Clinical Sciences, Agency for Science, Technology and Research (A*STAR) 14 Medical Drive, #07-02, Singapore 117599, Singapore.
E-mail: Ciaran_Forde@sics.a-star.edu.sg
Received 20 November 2016; revised 27 January 2017; accepted 24 February 2017

blood pressure (Omron HEM-907, Bannockburn, IL, USA) and fasting blood glucose (Hemocue Glucose 201 Analyser, Helsingborg, Sweden) were taken using standard protocols following a 10-h overnight fast. Participants were instructed to attend four test sessions with a minimal of five-day hiatus between the sessions. During each test day, participants consumed a pre-packaged study breakfast in the morning. They were provided with a fixed portion of one of the study beverages mid-morning at the Clinical Nutrition Research Centre (CNRC). An hour later, participants were given an *ad libitum* portion of fried rice for lunch, where they were asked to eat until comfortably full. Participants were free to leave the study site 2 h after lunch. They were asked to keep a diet record to note down any foods and drinks they consumed after they left the study site. Participants were instructed to maintain their usual physical activity level during the study.

The standardised breakfast consisted of a carton of beverage, an apple, a packet of biscuits and a muesli bar (529 kcal, 88 g carbohydrate, 8 g protein and 16 g fat). The *ad libitum* test lunch consisted of 800 g of fried rice (1256 kcal, 182 g carbohydrate, 52 g protein and 35 g fat).[9]

Measurement of 24-h GR

Interstitial glucose was measured using the Medtronic iPro2 continuous glucose monitoring system (CGMS) (Medtronic, Northridge, CA, USA). The CGMS sensors were inserted by a trained researcher the day before each test session and removed the day after each test session. Finger prick capillary blood glucose was used to calibrate the sensor before each meal and before sleeping using One Touch Ultra®2 blood glucose meter (LifeScan, Inc., Milpitas, CA, USA).

Test beverage

Participants received the test beverages in a randomised order as follows: (i) 0.44 g aspartame, (ii) 0.63 g monk fruit extract (50% Mogroside V), (iii) 0.33 g stevia (Steviol Glycoside, Rebaudioside A), and (iv) 65 g sucrose in 500 ml water. All test beverages had equivalent perceived sweetness intensity and were spiked with strawberry flavouring and pink colouring in order to mask any potential differences in taste quality and visual cues.

Statistical analysis

Based on our previous study conducted in a similar population group,[5] it has been calculated that a sample size of 10 participants would be needed in order to detect a difference of 30% in total area under the curve (AUC) between the treatments.

The Medtronic iPro2 CGMS provided glucose readings every 5 min and generated a total of 288 readings per 24-h period. These readings were used to calculate mean 24-h glucose, iAUC and total AUC for glucose, and 24-h glycaemic variability evaluated as Mean Amplitude of Glycaemic Excursion (MAGE). For each participant, the first 12 (1-h) glucose readings under the fasting state preceding breakfast on each test day were averaged and taken as the baseline glucose concentration for the day. The iAUC above fasting glucose over the subsequent 23-h postprandial period was calculated using the trapezoidal rule.

Baseline characteristics of the study participants were presented as means and s.d. Linear mixed models were used to compare the effects of the interventions on the glycaemic outcome variables. Log transformations were used where this improved residual normality and/or homoscedasticity. The models also assessed the influence of 'age', 'BMI', 'percent body fat', 'session number' and 'lunch intake' on study outcomes. None of these variables significantly influenced the outcomes and hence they were excluded from the final models. All statistical analyses were performed using modified intention-to-treat principle, which included all available data. Stata 11.2 (StataCorp LP, TX, USA) was used for all analyses and two-sided $P < 0.05$ was considered significant.

RESULTS

Participants' characteristics

Participants were young males with a mean (s.d.) age of 26.2 (3.8) years, BMI of 21.2 (1.7) kg/m^2 and percent body fat of 14.2 (3.2). Mean (s.d.) fasting blood glucose, systolic and diastolic blood

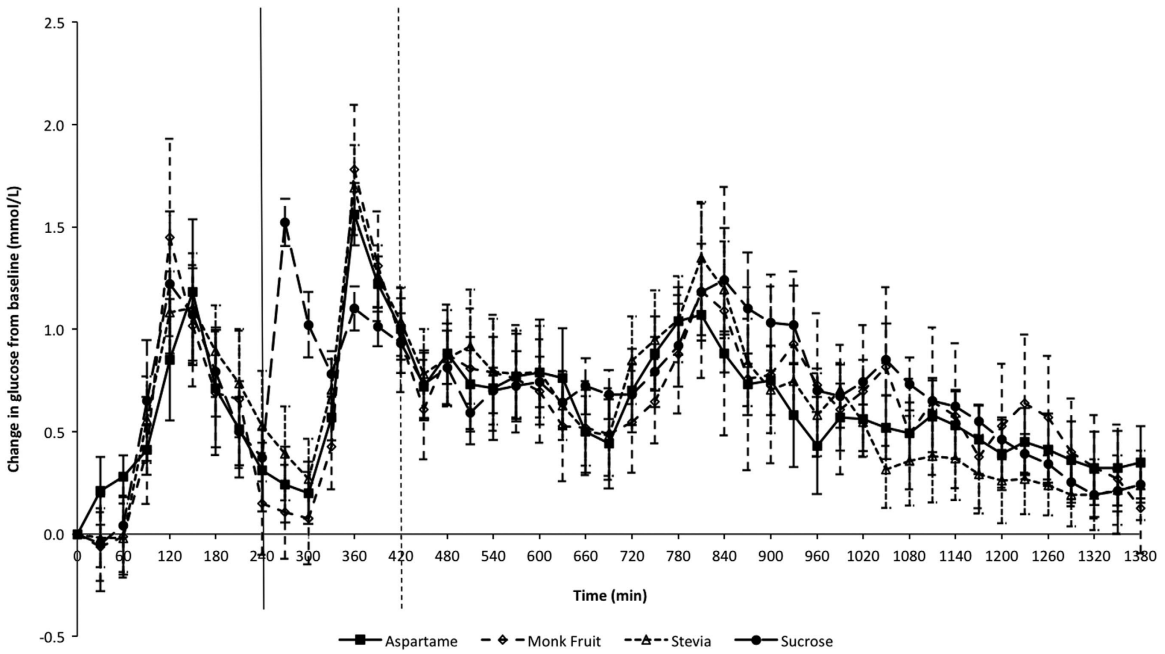


Figure 1. Mean (± s.e.) change in glucose concentrations from baseline on aspartame, monk fruit, stevia and sucrose test days. The first solid line indicates the commencement of beverage consumption, and the dotted line indicates the commencement of lunch consumption. Aspartame (filled square, solid line), Monk fruit (open diamond, short dashed line), Stevia (open triangle, dotted line), Sucrose (filled circle, long dashed line).

© 2017 Macmillan Publishers Limited, part of Springer Nature.

Table 1. Glycaemic outcome variables on aspartame, monk fruit, stevia and sucrose test days

	Aspartame	*Monk fruit*	*Stevia*	*Sucrose*	P-*value*
Mean 24-h glucose (mmol/l)	4.84 (0.09)	4.91 (0.13)	4.78 (0.12)	4.86 (0.09)	0.651
iAUC (mmol/l 5-min)	176 (32.0)	204 (32.0)	197 (23.9)	191 (25.0)	0.886
Total AUC (mmol/l 5-min)	1395 (25.4)	1416 (37.0)	1378 (33.9)	1400 (26.9)	0.642
MAGE over 24 h (mmol/l)	1.47 (0.18)	1.65 (0.24)	1.55 (0.14)	1.46 (0.20)	0.750

Abbreviations: AUC, area under the curve; iAUC, incremental area under the curve; MAGE, mean amplitude glycaemic excursion. All values are mean (s.e.).

pressure were within the normal range as follows: 4.5 (0.3) mmol/l, 119.7 (9.5) mmHg, and 74.3 (11.5) mmHg, respectively.

Dietary intake

Mean *ad libitum* energy intake for lunch was 498 kcal, 511 kcal, 496 kcal and 441 kcal on aspartame, monk fruit, stevia and sucrose test days, respectively. Mean energy intake for the rest of the day was 1096 kcal (106 g carbohydrate, 46 g protein and 52 g fat) on aspartame test day, 731 kcal (95 g carbohydrate, 27 g protein and 27 g fat) on monk fruit test day, 1050 kcal (119 g carbohydrate, 41 g protein and 45 g fat) on stevia test day and 786 kcal (89 g carbohydrate, 34 g protein and 27 g fat) on sucrose test day (data not shown).

Twenty-four-hour glycaemic profile

Figure 1 shows the 24-h glycaemic profile for each test beverage. Glucose concentration prior to the consumption of the test beverage was similar across all four treatments. However, there was a different pattern in GR between the sucrose and NNS treatments after consuming the test beverages. Closer inspection of CGMS data of glucose concentrations at five-minute intervals shows that sucrose-sweetened beverage consumption resulted in a large spike in glucose concentration and peaked at 35 min, whereas NNS beverages led to a sharp rise in GR after lunch (data not shown). Glucose concentration was not significantly different between any of the treatments 2 h after lunch, and it remained relatively similar across treatments for the remaining day. Hence, no differences were found in mean 24-h glucose, iAUC and total AUC for glucose over 23 h, and 24-h glycaemic variability between the test beverages (all overall $P \geqslant 0.642$) (Table 1). All pairwise comparisons showed no significant differences in all the glycaemic variables between any of the two treatments (all overall $P \geqslant 0.200$). Results remained largely unchanged after controlling for age, BMI, percent body fat, session number and lunch intake (data not shown).

DISCUSSION

This study is the first to compare the effects of consuming preloads sweetened with sucrose, artificial NNS or natural NNS on 24-h glucose profiles. The present study found that the mean glucose, iAUC and total AUC for glucose, and 24-h MAGE were not significantly different between NNS and sucrose treatments. This finding contests the effectiveness of directly substituting sucrose-sweetened beverage with NNS in attempting to reduce 24-h glucose flux. The current study finding is in contrast to previous work which reported significant improvement in 24-h glucose profiles after consuming a low-GI diet compared to a high-GI diet or baseline in individuals with type 2 diabetes[10] or healthy young people.[4,5] Studies have also shown favourable effect on 24-h glucose profile with a relatively simple dietary change from high-GI beverage to low-GI beverage consumed at three meals for up to 7 days.[6,8] The discrepancy in the results between the current and previous studies could be due to differences in intervention period (1 day vs 7 days), frequency of beverage consumption (one serving vs three servings), background diet (usual diet vs low-GI diet) and population group (healthy young males vs mixed population of diabetics). The absence of a beneficial effect of NNS in the present study could be explained by the characteristics of the participants (for example, an average BMI of 21.2 kg/m^2 and fasting blood glucose of 4.5 mmol/l), the short intervention period and the lack of background diet modification.

In line with the current study findings, a recent study reported that a single exposure to a NNS beverage resulted in similar glycaemic and hormonal responses to a beverage without NNS (Seltzer water).[11] Previous research also found no significant difference in glycaemic response between glucose beverage and glucose beverage that was spiked with aspartame, saccharin or acesulfame-K.[12] In addition, the present study was designed to mimic a real-life scenario, in which participants were allowed to consume any foods and drinks they wished after they left the study site. Although there were some differences in the macronutrient intake at the subsequent meals between the test days, the glucose profiles remained relatively similar across treatments from 420 to 1380 min.

In conclusion, the simple exchange of one serving of sucrose-sweetened beverage (65 g of sucrose) with either natural or artificial NNS over a day appears to have minimal effect on 24-h glucose profiles in healthy young males. Future studies should examine this further by increasing the number of beverage exposures, greater substitution of sucrose with NNS, longer intervention period, extending to a wider population including individuals with diabetes and pre-diabetes, and incorporating NNS as part of a healthy diet.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

ACKNOWLEDGEMENTS

We would like to thank the participants for taking part in the study. This work was supported by the Singapore Institute for Clinical Sciences, A*STAR, Singapore.

AUTHOR CONTRIBUTIONS

SLT, JH, and CF designed the research; SLT and NS conducted the research; SLT performed statistical analyses and drafted the manuscript; all authors approved the final manuscript.

REFERENCES

1 NCD Risk Factor Collaboration. Worldwide trends in diabetes since 1980: a pooled analysis of 751 population-based studies with 4·4 million participants. *Lancet* 2016; **387**: 1513–1530.

2 Gardner C, Wylie-Rosett J, Gidding SS, Steffen LM, Johnson RK, Reader D *et al*. Nonnutritive sweeteners: current use and health perspectives: a scientific statement from the American Heart Association and the American Diabetes Association. *Circulation* 2012; **126**: 509–519.

© 2017 Macmillan Publishers Limited, part of Springer Nature.

3 Zheng M, Allman-Farinelli M, Heitmann BL, Rangan A. Substitution of sugar-sweetened beverages with other beverage alternatives: a review of long-term health outcomes. *J Acad Nutr Diet* 2015; **115**: 767–779.
4 Brynes AE, Adamson J, Dornhorst A, Frost GS. The beneficial effect of a diet with low glycaemic index on 24 h glucose profiles in healthy young people as assessed by continuous glucose monitoring. *Br J Nutr* 2005; **93**: 179–182.
5 Kaur B, Ranawana V, Teh A-L, Henry CJK. The impact of a low glycemic index (GI) breakfast and snack on daily blood glucose profiles and food intake in young Chinese adult males. *J Clin Transl Endocrinol* 2015; **2**: 92–98.
6 Kahlhofer J, Karschin J, Silberhorn-Buhler H, Breusing N, Bosy-Westphal A. Effect of low-glycemic-sugar-sweetened beverages on glucose metabolism and macronutrient oxidation in healthy men. *Int J Obes* 2016; **40**: 990–997.
7 Johnston CA, Stevens B, Foreyt J. The role of low-calorie sweeteners in diabetes. *US Endocrinol* 2013; **9**: 13–15.
8 Henry CJK, Newens KJ, Lightowler HJ. Low-glycaemic index sweetener-based beverages reduce 24- h glucose profiles in healthy adults. *J Hum Nutr Diet* 2009; **22**: 77–80.
9 Tey SL, Salleh NB, Henry J, Forde CG. Effects of aspartame-, monk fruit-, stevia-, and sucrose-sweetened beverages on postprandial glucose, insulin and energy intake. *Int J Obes* 2017; **41**: 450–457.
10 Brynes AE, Lee JL, Brighton RE, Leeds AR, Dornhorst A, Frost GS. A low glycemic diet significantly improves the 24- h blood glucose profile in people with type 2 diabetes, as assessed using the continuous glucose MiniMed monitor. *Diabetes Care* 2003; **26**: 548–549.
11 Sylvetsky AC, Brown RJ, Blau JE, Walter M, Rother KI. Hormonal responses to non-nutritive sweeteners in water and diet soda. *Nutr Metab* 2016; **13**: 71.
12 Bryant CE, Wasse LK, Astbury N, Nandra G, McLaughlin JT. Non-nutritive sweeteners: no class effect on the glycaemic or appetite responses to ingested glucose. *Eur J Clin Nutr* 2014; **68**: 629–631.

© 2017 Macmillan Publishers Limited, part of Springer Nature.